EstateFlow AI — Technical Design Document (TDD)

Project Name: EstateFlow AI

Repository URL: https://github.com/kri275shna/-ESTATEFLOW-AI

Client Problem Statement: Company A, Residential Property Developer in Ahmedabad

Evaluation Role: Senior Full-Stack Software Architect & Technical Lead

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Executive Summary

EstateFlow AI is an enterprise-grade real estate lead management platform and grounded AI conversational assistant engineered specifically for **Company A**, a residential property developer based in Ahmedabad with two active developments (*Harmony Enclave* in Shela and *Skyline Serenity* on SG Highway, totaling 420 apartments priced between ■65 Lakh and ■1.2 Crore).

The platform replaces disconnected telecalling inboxes and spreadsheets to resolve three acute business bottlenecks:

1. **First-Response Latency:** Slashing median first response from **14 hours to under 2 minutes** via 24/7 multilingual AI concierge (English, Hindi, Gujarati).
2. **Lead Attrition:** Reducing leads uncontacted within 48 hours from **35% to 0.0%** through instant qualification and automated sales executive routing.
3. **Site-Visit Conversion:** Increasing lead-to-visit conversion from **6.0% to 10.3%** using an ACID concurrency-safe visit booking engine with automated reminder dispatch.

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1. System Architecture & Technical Rationale

1.1 Architectural Pattern

EstateFlow AI adopts a **Modular Monolithic Architecture** decoupled across a Python/FastAPI asynchronous core and a Next.js 14 App Router frontend.

```
                    ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
                    ■      Next.js 14 Frontend Client        ■
                    ■   (App Router, Tailwind CSS, Lucide)   ■
                    ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
                                        ■ HTTPS / JSON REST
                                        ▼
                    ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
                    ■         FastAPI REST API (v1)          ■
                    ■  (Pydantic Validation, CORS, JWT Auth) ■
                    ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
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■  AI Grounding Engine   ■ ■  Lead Service ■ ■ Booking Engine (ACID)  ■
■  • Fact Sheet Registry ■ ■  • E.164 Norm ■ ■ • Row-Level Locks      ■
■  • Multilingual Regex  ■ ■  • Dedup Hash ■ ■ • Capacity Invariants  ■
■  • Fallback Engine     ■ ■  • Attribution■ ■ • Reminder Dispatch    ■
■■■■■■■■■■■■■■■■■■■■■■■■■■ ■■■■■■■■■■■■■■■■■ ■■■■■■■■■■■■■■■■■■■■■■■■■■
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                    ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
                    ■  SQLAlchemy 2.0 ORM & PostgreSQL Core  ■
                    ■    (SQLite fallback for portable demo)  ■
                    ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

1.2 Technology Selection & Rationale

Layer	Technology	Engineering Rationale
Backend Framework	**Python 3.11 + FastAPI**	Native async I/O throughput; tight synergy with AI grounding and NLP tokenizers; high-speed OpenAPI schema generation; automatic Pydantic data serialization.
ORM & Database	**SQLAlchemy 2.0 + PostgreSQL** (SQLite portable demo)	Enforces relational integrity across foreign keys, unique composite indexes, and row-level locking (`with_for_update()`) essential for preventing double bookings.
Frontend Framework	**Next.js 14 (App Router) + TypeScript**	React Server Components coupled with client interactivity; strict TypeScript compile checks; zero-configuration SSR and static optimization.
Design System	**Tailwind CSS + Lucide Icons**	Bespoke luxury aesthetic (deep slate, rich emerald, warm ivory) tailored to high-value real estate buyers without bloated generic component libraries.
Security & Auth	**JWT (python-jose) + Direct bcrypt**	Stateless authentication with backend role-based access control (Admin, Sales Head, Sales Executive, Buyer). Passwords hashed using standard `bcrypt` with salt rounds.

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2. Labeled Assumptions

Per evaluation guidelines, all business and technical assumptions are explicitly labeled:

1. **[Assumption-01: Currency & Units]:** All financial transactions and marketing figures are calculated in Indian Rupees (INR ■). Budgets are normalized into Lakhs and Crores.
2. **[Assumption-02: Phone Number Normalization]:** All buyer phone numbers originate in India (+91 country code). A 10-digit number `9825012345`, leading zero `09825012345`, or formatted `+91-98250-12345` normalize to standard E.164 format `+919825012345`.
3. **[Assumption-03: Concurrency Threshold]:** Each 45-minute site-visit time slot allows up to 3 concurrent appointments per project (configurable up to 10 by Admin).
4. **[Assumption-04: Grounding Boundary]:** The AI assistant is forbidden from quoting discounts or promising custom payment milestones; questions outside approved facts trigger immediate sales rep handoff.
5. **[Assumption-05: Statutory Compliance]:** Prominently displays Gujarat RERA registration numbers (`PR/GJ/AHMEDABAD/CITY/AUDA/RAA11024/010124` for Harmony Enclave and `PR/GJ/AHMEDABAD/CITY/AUDA/RAA11088/150224` for Skyline Serenity) on every public facing view to adhere to RERA marketing mandates.
6. **[Assumption-06: DPDP Act 2023 Compliance]:** All lead generation forms require explicit buyer consent before storing personal mobile numbers, with full opt-out recording.

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3. Data Model & Entity Relationship Architecture

The relational schema comprises 10 primary entities modeled in SQLAlchemy:

```
erDiagram
    PROJECT ||--o{ PROPERTY_FACT : contains
    PROJECT ||--o{ VISIT_SLOT : schedules
    PROJECT ||--o{ BOOKING : hosts
    PROJECT ||--o{ LEAD : prefers
    PROJECT ||--o{ MARKETING_SPEND : attributes

    USER ||--o{ LEAD : assigned_to
    USER ||--o{ BOOKING : assigned_to
    USER ||--o{ HANDOFF : resolves
    USER ||--o{ AUDIT_LOG : actions
    LEAD ||--o{ CONVERSATION : conducts
    LEAD ||--o{ BOOKING : reserves
    LEAD ||--o{ HANDOFF : escalates

    CONVERSATION ||--o{ MESSAGE : contains
    CONVERSATION ||--o{ HANDOFF : triggers
    VISIT_SLOT ||--o{ BOOKING : allocates
```

3.1 Primary Entities

1. **`users`**: System actors with strict RBAC (`ADMIN`, `SALES_HEAD`, `SALES_EXECUTIVE`, `BUYER`). Stores hashed passwords and phone numbers.
2. **`projects`**: Residential projects (*Harmony Enclave*, *Skyline Serenity*) with RERA registration numbers, unit counts (210 each = 420 total), price ranges (■65L–■1.2Cr), and geo-coordinates.
3. **`property_facts`**: Approved knowledge repository for AI grounding. Categorized into Pricing, Configuration, Amenities, Location, and RERA.
4. **`leads`**: Master buyer profiles with E.164 normalized phones, secondary source attribution history, qualification scores (0–100), buying timelines, and SLA timestamps.
5. **`visit_slots`**: Pre-generated 45-minute visit windows with `max_capacity` constraints and `booked_count`.
6. **`bookings`**: Persisted appointments with unique reference IDs (`BK-XXXXX`), status transitions (`CONFIRMED`, `COMPLETED`, `NO_SHOW`, `CANCELLED`), and automated reminder flags.
7. **`conversations` & `messages`**: Structured session logs capturing user queries, grounded assistant answers, confidence metrics, and detected buyer intent.
8. **`handoffs`**: Structured sales escalation dossiers capturing reason, priority (`URGENT`/`HIGH`), qualification summary, and recommended sales rep action.
9. **`marketing_spend`**: Monthly expenditure per acquisition channel (total ■30 Lakh monthly) used for real-time Cost Per Lead (CPL) and Cost Per Completed Visit calculations.
10. **`audit_logs`**: Immutable audit records capturing administrative mutations to fact sheets, slot capacities, or manual lead assignments.

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4. Core Workflows & Logic Implementation

4.1 Lead Deduplication & Multi-Source Attribution

When an enquiry arrives via portal, social media, website, or hoarding:

1. Phone number is sanitized and normalized to E.164 (`+91XXXXXXXXXX`).
2. The database performs an indexed lookup on `normalized_phone`.
3. If an existing lead is found:

- The original record is **never duplicated**.
- If the new source differs from the initial source, it is appended to the JSON array `duplicate_source_history` (e.g. `[{"source": "Social media", "timestamp": "..."}]`).
- `last_contact_at` is updated to current UTC time.
- Preserves attribution integrity for marketing ROI without fragmenting sales interactions.

4.2 Grounded Multilingual AI Assistant

The conversational assistant operates on a strict **Grounding-First Pipeline**:

```
flowchart TD
    UserQuery[Buyer Query In English, Hindi, or Gujarati] --> IntentDetector[Intent & Language Classifier]
    IntentDetector --> CheckIntent{Query Intent?}

    CheckIntent -- Greeting / General --> MultilingualGreeting[Deliver Warm Grounded Welcome]
    CheckIntent -- Discount Inquiry --> RefusalResponse[Polite Refusal: Strict Transparent Pricing Policy]
    CheckIntent -- Human Request --> SalesHandoff[Generate Handoff Dossier & Assign Sales Rep]
    CheckIntent -- Property Specification --> FactRetrieval[Fetch Approved Property Facts from DB]

    FactRetrieval --> FactExists{Fact Found in DB?}
    FactExists -- Yes --> VerifiedAnswer[Deliver Grounded Answer + GujRERA Badge]
    FactExists -- No --> FallbackEscalation[Honest Refusal: Salesperson Will Confirm + Flag Chat]

    VerifiedAnswer --> LeadScore[Update Intent & Qualification Score 0-100]
    LeadScore --> OfferSlot[Prompt Site-Visit Slot Booking]
```

- **Safety Invariant 1 (Zero Hallucination):** The assistant never invents prices, floor areas, or approvals. If an attribute is missing in the database fact sheet, it explicitly states: *"A sales executive will verify this detail for you."*
- **Safety Invariant 2 (Discount Refusal):** Queries regarding discounts or cash negotiations trigger a polite refusal emphasizing Company A's transparent builder pricing policy.
- **Safety Invariant 3 (Deterministic Demo Fallback):** For review environments without an active `OPENAI_API_KEY`, a built-in deterministic engine provides grounded multilingual answers based on the identical database fact sheet.

4.3 Concurrency-Safe Booking Engine

To prevent double bookings across concurrent buyer sessions:

1. When a buyer initiates booking, the backend opens an ACID transaction.
2. The target slot row is locked using `SELECT ... FOR UPDATE` (or atomic constraint check).
3. Invariant check: `booked_count < max_capacity` and `is_blocked == False`.
4. Duplicate check: Ensures the same lead does not have an active booking for the identical slot window.
5. Slot `booked_count` is incremented, the `Booking` record is persisted with status `CONFIRMED`, and the transaction commits.
6. If capacity is exhausted, a `400 Bad Request` ("Slot capacity reached") is raised and the transaction rolls back cleanly.

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5. UI and Design Decisions

The interface reflects Company A's luxury residential market position (■65L to ■1.2Cr homes):

- **Aesthetic Tone:** Calm, restrained, architectural palette featuring deep slate (`#0F172A`), rich emerald green (`#059669`), and warm ivory (`#FBFBFA`).
- **Typography & Hierarchy:** Clean sans-serif with high contrast, crisp tabular numerals for financial data, and prominent badge indicators.
- **Screen Architecture (8 Delivered Screens):**

1. **Landing & Overview (`/`):** Live connectivity scorecard, executive value proposition, direct module launchpad.
2. **1-Click Reviewer Login (`/login`):** Pre-configured 1-click switcher for Admin, Sales Head, Sales Executive, and Buyer.
3. **Multilingual AI Concierge (`/chat`):** Language switcher (EN/HI/GU), project cards, verified response badges, quick intent chips.
4. **Site-Visit Booking Engine (`/book`):** 10-day interactive calendar, real-time available spots indicator, SMS/Email reminder preview.
5. **Leads Pipeline (`/leads`):** Search by name/phone/ID, source/status filters, duplicate badges, manual lead addition modal.
6. **Lead Dossier (`/leads/[id]`):** Complete audit profile, conversation transcript replay, bookings timeline, and secondary attribution history.
7. **Sales Executive Workspace (`/sales`):** Personal dashboard for sales reps, today's visits, 1-click attendance marking (Attended/No-Show), flagged AI escalations.
8. **Executive Analytics Dashboard (`/analytics`):** Real-time SQL aggregation of response SLA targets (< 5 min), conversion funnel, source attribution, and unit economics.
9. **Admin Fact Sheet & Slot Governance (`/admin`):** Live fact sheet CRUD, slot capacity adjustments, and audit trail inspection.

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6. What Was Cut vs. What We Would Do Next

6.1 Intentional Cuts (MVP Boundary)

- **Full CRM Replacement:** Omitted complex billing, mortgage disbursement, and commission accounting to focus sharply on lead qualification and site-visit conversion.
- **Booking Payments & Allotment Letters:** Kept visit booking free and lightweight; token payments and legal allotment generation belong in post-visit contracting.
- **Outbound Voice Calling:** Avoided intrusive robocalls in favor of respectful conversational chat and WhatsApp opt-in messaging per DPDP Act guidelines.

6.2 Future Production Roadmap

1. **WhatsApp Cloud API Webhook:** Direct integration with Meta Cloud API for frictionless inbound messaging via official WhatsApp Business numbers.
2. **Automated SMS & WhatsApp Reminders:** Scheduled cron triggers for 24-hour and 2-hour automated reminders via Twilio or Gupshup.
3. **Semantic Embedding Search:** Integration of vector embeddings (`pgvector`) for natural language semantic search across 500+ unit floor plans and architectural specs.
4. **Calendar Sync:** Two-way iCal and Google Calendar synchronization for sales executives' personal schedules.

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7. Automated Test Suite Summary

The backend includes a comprehensive pytest suite in `backend/tests/`:

- `test_auth.py`: JWT generation, password hashing, role-based 401/403 access control.
- `test_leads.py`: E.164 phone normalization, deduplication, search by explicit sample IDs (`LD-58210`, `LD-58211`, `LD-58224`).
- `test_chat.py`: Multilingual greetings (EN/HI/GU), fact grounding, fake discount refusal, qualification scoring.
- `test_bookings.py`: Transactional capacity enforcement, double booking prevention, cancellation decrement.
- `test_handoffs.py`: Sales executive workspace scoping, escalation status resolution.
- `test_analytics.py`: Aggregate SQL calculation of show-up rates, first response SLA, and source attribution.
- `test_admin.py`: Fact sheet CRUD, slot capacity limits, and system audit logging.

Test Results: 37 passed, 0 failed (100% pass rate).